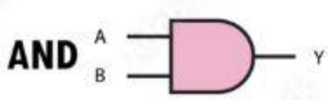
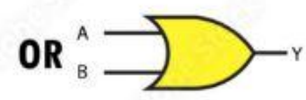
My First Logic Gate

Cassian needs to make sure his compound is locked up. He has two gates that allow people in, but only if they're opened properly. The first gate uses AND logic - it will only turn on (1) if both of its inputs are on. The second uses OR logic - it will turn on (1) if either of its inputs is on. Help him test his gates by showing the output for both. (This is a terrible security setup, but it's a start.)

Cassian's configuration is similar to a computer's logic gates. Logic gates are computer components that take one or more binary inputs (0 or 1) and produce a binary output (0 or 1). Their logic can be represented in a truth table. A truth table shows what the outputs should be for each input configuration. Below are two examples of logic gates and their corresponding truth table.



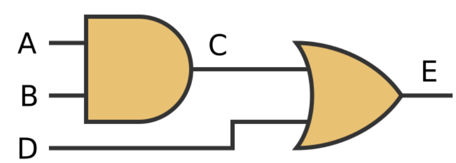
A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1



A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

An **AND** gate has an output that is a 1 if both the inputs are 1. An **OR** gate has an output that is 1 if either of the inputs are 1.

Gates can be put in place sequentially into a circuit, where the output of one gate is the input of another gate. The diagram above shows a sequential circuit with an AND gate and an OR gate. Your goal is to calculate the intermediate signal (C) and the output of this circuit (E) given the inputs (A, B, D).



Input

You will read 3 bits (A B D) on one line.

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1 0 0
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Output

You will output 2 bits. Print the intermediate signal (C), then the output of the given circuit (E). Make sure they are in order (C E).

```
0 0
```

Discussion

HINT: First calculate the intermediate signal (C) from the AND gate, then use it as an input to the OR gate. Most coding languages have a built in variable type for bit, like there is for int. They also have built in operators for AND (&) as well as OR (|).

Reminder: have you run your solution against **all** of the student data sets?

Additional Examples

Input 2	Output 2	Input 3	Output 3
1 0 1	0 1	1 1 0	1 1